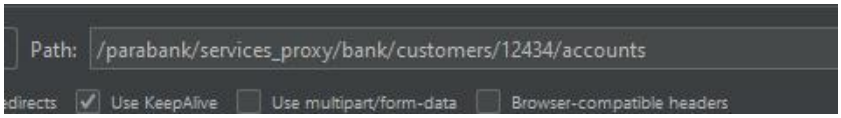
Test Design

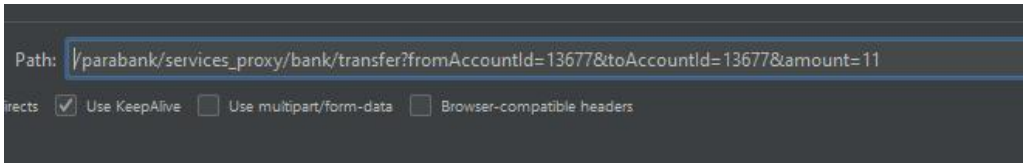
The test plan for the Parabank website consisted of 11 HTTP request scripts that were obtained and chosen from an HTTP(S) Test Script recorder session. These requests include:

1. Accessing the Homepage in /index.htm
2. Login with the username chaelg11! and password abc1234
3. Accessing account overview
4. Redirected to overview.htm
5. Clicking account number for account details
6. Redirected to activity.htm
7. Clicking transfer funds
8. Transferring 11 dollars from and to the user's own bank account
9. Clicking find transactions
10. Finding transactions by amount: 11 dollars
11. Clicking logout.

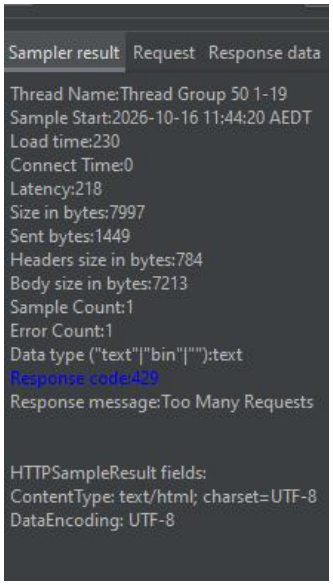
Due to log in using chaelg11! as a username and abc1234 as password, this account needs to be created before running the tests. Parabank has a database reset after a certain amount of times therefore before every test run, log in was tested manually and account needs to be registered if the database has been reset.

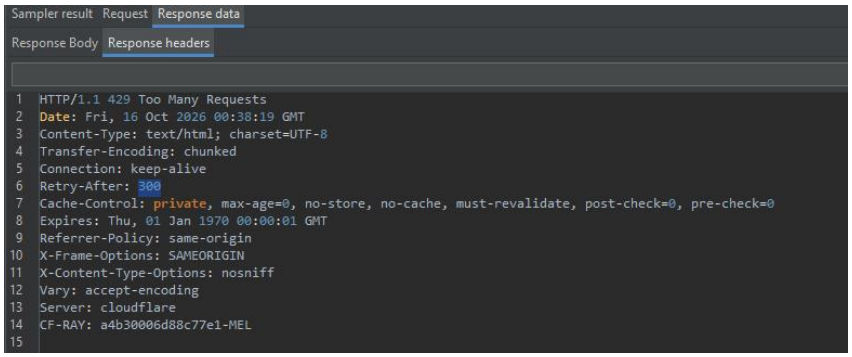


Additionally, clicking account overview and clicking account number for account details has the customer ID hardcoded in the path, therefore this customer ID needs to be acquired. In this test, this customer ID was acquired through the HTTP Script Recorder.



Similarly, the account number is also hardcoded in the path for transferring funds and finding transactions. The method to acquire this account number is simpler as this account number can be obtained in the account overview.

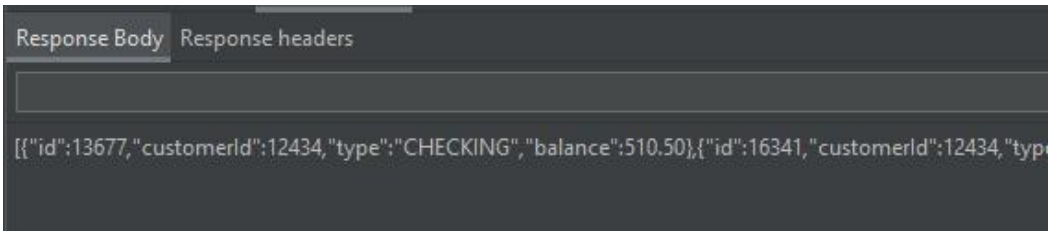




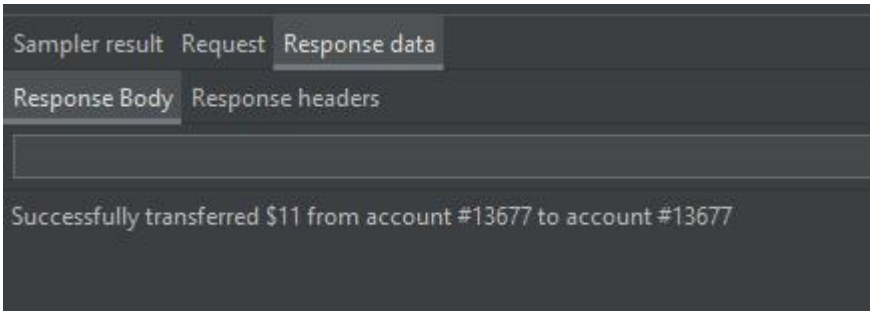
During initial test runs, a significant (>80% error%) request failure was observed for thread group with 10 users or more. This was determined to be due to rate limiting features of the Parabank website preventing an IP address from sending a large number of requests with response code 429 observed in the sampler result in the results tree. The retry after line in response header shows a count of 300 seconds meaning after the first 429 error, any HTTP request from the IP address is blocked for 5 minutes.

In the interest of writing this report, the test plan implements timer delay before each HTTP request to combat this rate limit in the website. Simple requests such as clicking and redirecting used a 1-2 seconds timer while requests which includes filling an input field such as log in and transferring funds used 1-3 seconds timer to mimic the delay of a human user. This fully prevented code 429 in the 10 user group but did not fully prevent code 429 in the 20 and 50 user groups. However, due to the test duration being 20 seconds, it was decided that increasing the delay further would hinder test runs especially for the 1 and 10 user group.

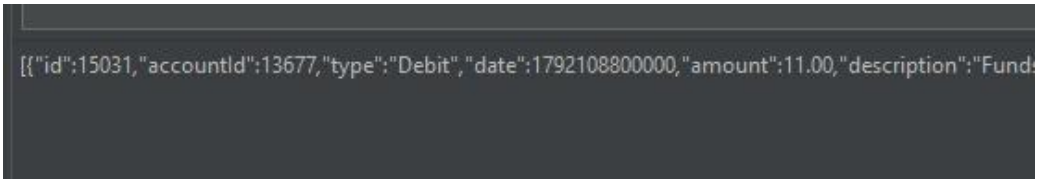
Most of the clicking and redirecting requests does not contain many useful response data therefore assertions for these requests will be simply asserting response code 200 for successful requests.



For account overview and account detail request, the response body contains information such as account number, customerId, type of account, and balance. For this request a text response assertion matching pattern that contains “balance”, “customerId”, and “id” is used on top of response code 200. The specific balance and ID values are determined to be unusable since they change for every transactions and for every account.



Response body for transferring 11 dollars are shown in the image above. Therefore the text response assertion used for this request includes matching pattern substring "Successfully transferred $11". Substring is used instead of contains as the $ symbol was determined to hinder the contains pattern matching.



Lastly, finding transactions by amount 11 dollars contain the response body which has the line "amount":11.00 which changes depending on what amount the search is used with. As the test always uses 11 dollars to search the text response for this request will be pattern matching that contains "amount":11. Other HTTP requests that were not mentioned having text response assertion uses a simple response code 200 assertion instead.

Results and Analysis

The performance testing was done using four thread groups configured with 1, 10, 20, and 50 users each with 10 second ramp up period and 20 second duration. The results were collected using JMeter Aggregate Report listener. With summarised overall findings as follows:

Metric	1 user	10 users	20 users	50 users
Total Samples	12	88	168	433
Throughput (requests/sec)	0.66	4.4	8.6	22
Average Response Time (ms)	253	247	141	53
Minimum Response Time (ms)	193	189	8	6
Maximum Response Time (ms)	447	488	673	691
95th Percentile (ms)	414	427	422	318
Error %	0.00%	0.00%	47.02%	85.22%

Table 1. Summary of Aggregate Reports from all request from the 4 thread group

Metric	1 User	10 Users	20 Users	50 Users
Home Page				
Samples	2	11	22	58
Throughput (req/sec)	0.11	0.62	1.1	3
Average (ms)	206	217	200	144
Min (ms)	197	191	9	8
Max (ms)	215	242	354	691
95th Percentile (ms)	215	236	228	318
Error %	0.00%	0.00%	9.09%	44.83%
Login				
Samples	1	10	20	50
Throughput (req/sec)	2.4	1	1.9	4.9
Average (ms)	414	418	433	177
Min (ms)	414	408	271	8
Max (ms)	414	430	673	536
95th Percentile (ms)	414	427	562	436
Error %	0.00%	0.00%	5.00%	62.00%

Table 2. Summary of Aggregate Reports for Homepage and Login request from the 4 thread group

*Raw aggregate report tables are shown in the appendix

Based on Table 1, The throughput increased consistently as the number of concurrent users increased. The 1 user Thread Group recorded approximately 0.66 requests per second, increasing to 4.4 requests per second for 10 users, 8.6 for 20 users, and 22 for 50 users. This indicates that more HTTP requests were completed as the number of concurrent users increased. However, throughput measures both successful and unsuccessful requests. Therefore, the higher throughput observed in the 20 and 50 user Thread Groups does not necessarily indicate that the website successfully processed more user operations.

The 1 and 10 user Thread Groups demonstrated relatively stable performance with no recorded errors. The average response time decreased slightly from 253 ms to 247 ms, while the maximum response time increased from 447 ms to 488 ms. The 95th percentile response time also remained relatively consistent at 414 ms and 427 ms respectively. This indicates that increasing the number of concurrent users from 1 to 10 did not significantly affect the response times or reliability of the tested requests.

However, a noticeable change was observed in the 20 and50 user Thread Groups. The error percentage increased significantly to 47.02% and 85.22% respectively. Despite this, the average response time decreased to 141 ms for 20 users and 53 ms for 50 users. This reduction is not associated with improved performance, as many requests were unsuccessful due to HTTP 429 responses. These responses may be rejected quickly without completing the intended operation, reducing the recorded minimum and consequently average response time.

Table 2 provides additional insights to this observation as Table 2 compiles results only for homepage and login request in which most of the requests were successful for the 20 user Thread Groups as these were the first 2 requests to be performed by any user. Table 2

shows that for the homepage request, the average response time remained relatively consistent across the first three workloads, at 206 ms, 217 ms, and 200 ms respectively. The error percentage remained at 0% for the 1 and 10 user groups before increasing to 9.09% at 20 users and 44.83% at 50 users. The maximum response time also increased slightly to 354 at 20 user groups and significantly to 691 ms at 50 users implying some requests took longer to process at these higher workloads.

Similarly, the login request recorded average response times of 414 ms, 418 ms, and 433 ms for 1, 10, and 20 users respectively. The error percentage remained at 0% for the first two workloads and increased only slightly to 5% at 20 users. However, the 95th percentile response time increased to 562 ms at 20 users compared with 427 ms at 10 users as well as the maximum response time increasing from 430 ms to 673 ms for 20 users suggesting that some login requests experienced longer response times.

The most significant issue observed during testing was the increase in error percentage at higher user loads. Both the 1 and 10 user groups recorded 0% errors, whereas the 20 user group recorded 47.02% and the 50 user group recorded 85.22%.

As discussed earlier in test design, HTTP 429 (Too Many Requests) responses were observed during testing. These responses indicate that the server or an associated protection mechanism was limiting incoming requests. The response headers also included a retry after value of 300 seconds, indicating that the client should wait 5 minutes before attempting another request. Due to this rate limiting feature, later requests in the 20 and 50 user groups were significantly affected by the 429 error response code although few initial requests, especially some of the homepage and login request, could be performed successfully.

While these results demonstrate that the public ParaBank website became increasingly unreliable under the tested workloads, this is only true for concurrent users that are sourced from the same IP address, it is likely this rate limiting feature could be addressed by setting up virtual users with varying IP address.

Conclusion

Overall, the performance testing demonstrated that ParaBank handled the 1 and 10 user workloads with relatively stable response times and no recorded errors. However, increasing concurrency to 20 and 50 users resulted in significant increases in request failures, reaching 47.02% and 85.22% respectively heavily due to rate limiting feature which was indicated by HTTP 429 responses. Additionally, a 20 seconds test duration limit including a 10 ramp up period limits the findings to the relatively short duration performance testing. Further testing using a controlled environment without rate limiting restrictions or set up with virtual users having varying IP address to attempt to workaround the rate limiting restrictions may be useful in further assessing the application’s performance more accurately.

Appendix

1 User aggregate report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Parabank hoempa...	2	206	197	215	215	215	197	215	0.00%	6.6/min	0.18	0.05
Parabank Login	1	414	414	414	414	414	414	414	0.00%	2.4/sec	5.61	3.42
Clicking account o...	1	196	196	196	196	196	196	196	0.00%	5.1/sec	1.85	2.64
Redirected to over...	1	447	447	447	447	447	447	447	0.00%	2.2/sec	4.55	1.31
Clicking account n...	1	195	195	195	195	195	195	195	0.00%	5.1/sec	1.86	2.65
Redirected to activ...	1	196	196	196	196	196	196	196	0.00%	5.1/sec	13.95	3.02
Clicking transfer f...	1	200	200	200	200	200	200	200	0.00%	5.0/sec	12.15	2.96
transferring funds ...	1	205	205	205	205	205	205	205	0.00%	4.9/sec	1.61	3.43
Clicking find trans...	1	198	198	198	198	198	198	198	0.00%	5.1/sec	15.03	2.95
Finding transactio...	1	193	193	193	193	193	193	193	0.00%	5.2/sec	2.85	2.89
Clicking logout	1	388	388	388	388	388	388	388	0.00%	2.6/sec	4.87	3.04
TOTAL	12	253	198	414	414	447	193	447	0.00%	39.5/min	1.06	0.45

10 User aggregate report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Parabank hoempa...	11	217	215	236	236	242	191	242	0.00%	37.4/min	1.05	0.29
Parabank Login	10	418	418	427	427	430	408	430	0.00%	59.9/min	2.44	1.41
Clicking account o...	10	193	194	196	196	197	189	197	0.00%	1.1/sec	0.38	0.54
Redirected to over...	10	240	195	319	319	422	190	422	0.00%	1.0/sec	2.05	0.59
Clicking account n...	10	231	192	267	267	488	189	488	0.00%	1.0/sec	0.50	0.53
Redirected to activ...	10	229	204	245	245	435	192	435	0.00%	1.0/sec	2.89	0.60
Clicking transfer f...	9	235	207	247	467	467	192	467	0.00%	56.9/min	2.43	0.56
transferring funds ...	7	197	195	199	209	209	193	209	0.00%	50.6/min	0.28	0.59
Clicking find trans...	5	233	204	275	300	300	192	300	0.00%	49.4/min	2.45	0.48
Finding transactio...	4	193	193	195	195	195	191	195	0.00%	52.2/min	0.50	0.49
Clicking logout	2	384	382	387	387	387	382	387	0.00%	45.3/min	1.43	0.89
TOTAL	88	247	199	418	427	467	189	488	0.00%	4.4/sec	7.38	2.97

20 User aggregate report

Label	# Samples ↓	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Parabank hoempa...	22	200	210	223	228	354	9	354	9.09%	1.1/sec	2.61	0.5
Parabank Login	20	433	413	490	562	673	271	673	5.00%	1.9/sec	5.06	2.6
Clicking account o...	20	165	192	195	196	199	10	199	15.00%	1.8/sec	2.59	0.9
Redirected to over...	20	141	190	195	196	259	10	259	30.00%	1.8/sec	6.65	1.0
Clicking account n...	20	92	12	194	194	197	8	197	55.00%	1.7/sec	7.35	0.8
Redirected to activ...	19	79	12	195	201	369	10	369	68.42%	1.8/sec	10.85	1.0
Clicking transfer f...	18	52	12	191	194	201	9	201	77.78%	2.1/sec	13.39	1.2
transferring funds ...	13	10	10	13	13	14	9	14	100.00%	2.0/sec	15.23	1.4
Clicking find trans...	10	10	11	12	12	12	9	12	100.00%	2.2/sec	16.90	1.2
Finding transactio...	4	10	11	11	11	11	9	11	100.00%	1.2/sec	9.67	0.6
Clicking logout	2	10	9	12	12	12	9	12	100.00%	1.2/sec	9.39	0.7
TOTAL	168	141	190	404	422	490	8	673	47.02%	8.6/sec	38.28	5.6

50 User aggregate report

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
Parabank hoempa...	58	144	204	223	318	490	8	691	44.83%	3.0/sec	13.02	1.37
Parabank Login	50	177	12	427	436	536	8	536	62.00%	4.9/sec	27.38	4.87
Clicking account o...	50	50	11	192	194	203	8	203	78.00%	4.9/sec	29.47	2.45
Redirected to over...	50	17	10	13	15	191	7	191	96.00%	4.8/sec	33.32	2.72
Clicking account n...	50	10	10	12	12	15	8	15	100.00%	5.0/sec	37.99	2.50
Redirected to activ...	50	10	10	12	12	15	6	15	100.00%	5.3/sec	40.20	3.05
Clicking transfer f...	48	10	10	12	13	15	8	15	100.00%	5.3/sec	40.51	3.07
transferring funds ...	33	10	10	12	13	18	8	18	100.00%	4.4/sec	33.14	3.04
Clicking find trans...	21	9	10	11	13	13	8	13	100.00%	3.3/sec	24.95	1.93
Finding transactio...	14	10	10	12	12	14	8	14	100.00%	4.1/sec	30.82	2.29
Clicking logout	9	9	9	12	12	12	7	12	100.00%	5.1/sec	38.53	2.99
TOTAL	433	53	10	207	318	456	6	691	85.22%	22.0/sec	147.59	13.18